Final Review Solutions

Written by Katherine Miner (TA)

December 8th, 2025

Gronwall Inequality

Problem

Let $y : [0, T] \to \mathbb{R}$ satisfy

$$y'(t) \leq 3y(t) + 2, \qquad y(0) = y_0.$$

1. Use an integrating factor to derive an explicit upper bound of the form

$$y(t) \leq C_1 e^{3t} + C_2$$

for all $t \in [0, T]$, with C_1, C_2 in terms of y_0.

2. Re-derive the same bound using the integral inequality

$$y(t) \leq y_0 + \int_0^t \big(3y(s) + 2\big)\, ds$$

together with Grönwall's inequality.

Solution. **(a) Integrating factor bound.** We start from

$$y'(t) \leq 3y(t) + 2.$$

Rewrite as

$$y'(t) - 3y(t) \leq 2.$$

Multiply both sides by the integrating factor e^{-3t}:

$$e^{-3t}y'(t) - 3e^{-3t}y(t) \leq 2e^{-3t} \quad \Longrightarrow \quad \frac{d}{dt}\big(e^{-3t}y(t)\big) \leq 2e^{-3t}.$$

Integrate from 0 to t:

$$e^{-3t}y(t) - y_0 \leq \int_0^t 2e^{-3s}\, ds = 2\left[-\frac{1}{3}e^{-3s}\right]_0^t = \frac{2}{3}\big(1 - e^{-3t}\big).$$

Thus

$$e^{-3t}y(t) \leq y_0 + \frac{2}{3}\big(1 - e^{-3t}\big) = y_0 + \frac{2}{3} - \frac{2}{3}e^{-3t}.$$

Multiply by e^{3t}:

$$y(t) \leq e^{3t}\Big(y_0 + \frac{2}{3}\Big) - \frac{2}{3}.$$

So an explicit bound of the desired form is

$$y(t) \le C_1 e^{3t} + C_2, \quad \text{with } C_1 = y_0 + \frac{2}{3},\ C_2 = -\frac{2}{3}.$$

(b) Using Grönwall's inequality. We are given

$$y(t) \le y_0 + \int_0^t \big(3y(s) + 2\big)\, ds = y_0 + 2t + 3\int_0^t y(s)\, ds.$$

Define

$$w(t) := y(t) + \frac{2}{3}.$$

Then $y(t) = w(t) - \frac{2}{3}$, and

$$y_0 = w(0) - \frac{2}{3}.$$

Plug into the inequality:

$$\begin{aligned} w(t) - \frac{2}{3} &\le w(0) - \frac{2}{3} + 2t + 3\int_0^t \big(w(s) - \tfrac{2}{3}\big)\, ds \\ &= w(0) - \frac{2}{3} + 2t + 3\int_0^t w(s)\, ds - 2t. \end{aligned}$$

The $2t$ terms cancel, so

$$w(t) \le w(0) + 3\int_0^t w(s)\, ds.$$

This is exactly the hypothesis of Grönwall's inequality with

$$w(t) \le A + \int_0^t b\, w(s)\, ds, \quad A = w(0),\ b = 3.$$

Hence

$$w(t) \le w(0)\, e^{3t} = \Big(y_0 + \frac{2}{3}\Big) e^{3t}.$$

Therefore

$$y(t) = w(t) - \frac{2}{3} \le \Big(y_0 + \frac{2}{3}\Big) e^{3t} - \frac{2}{3},$$

which is exactly the same bound as in part (a). □

Comparison Principle

Problem

Consider

$$y'(t) \le 2t\,y(t) + t^2, \qquad y(0) = 0,$$

and the linear ODE

$$x'(t) = 2t\,x(t) + t^2, \qquad x(0) = 0.$$

1. Solve for $x(t)$ explicitly.
2. Use the comparison principle to show $y(t) \le x(t)$ for all $t \ge 0$.
3. Check that this upper bound is the same as what you obtain by directly applying Grönwall's inequality to the inequality for y.

Solution. **(a) Solve the linear ODE for $x(t)$.** We have

$$x'(t) = 2t\,x(t) + t^2, \qquad x(0) = 0.$$

Rewrite as

$$x'(t) - 2t\,x(t) = t^2.$$

The integrating factor is

$$\mu(t) = e^{-\int 2t\,dt} = e^{-t^2}.$$

Then

$$\frac{d}{dt}\big(e^{-t^2}x(t)\big) = t^2 e^{-t^2}.$$

Integrate from 0 to t:

$$e^{-t^2}x(t) - e^0 x(0) = \int_0^t s^2 e^{-s^2}\,ds.$$

Since $x(0) = 0$, we obtain

$$e^{-t^2}x(t) = \int_0^t s^2 e^{-s^2}\,ds \quad\Longrightarrow\quad x(t) = e^{t^2}\int_0^t s^2 e^{-s^2}\,ds.$$

(b) Comparison $y \le x$. We are given

$$y'(t) \le 2t\,y(t) + t^2, \qquad y(0) = 0.$$

Define

$$w(t) := x(t) - y(t).$$

Then

$$w(0) = x(0) - y(0) = 0.$$

Differentiate:

$$w'(t) = x'(t) - y'(t) \ge \big(2t\,x(t) + t^2\big) - \big(2t\,y(t) + t^2\big) = 2t\big(x(t) - y(t)\big) = 2t\,w(t).$$

Thus w satisfies

$$w'(t) - 2t\, w(t) \geq 0, \qquad w(0) = 0.$$

Consider the linear ODE

$$z'(t) - 2t\, z(t) = 0, \qquad z(0) = 0.$$

The unique solution is $z(t) \equiv 0$. By the comparison principle for scalar linear equations, we have $w(t) \geq z(t) = 0$ for all $t \geq 0$, i.e.

$$x(t) - y(t) = w(t) \geq 0 \quad \Longrightarrow \quad y(t) \leq x(t) \quad \text{for all } t \geq 0.$$

(c) Grönwall on y gives the same bound. From

$$y'(t) \leq 2t\, y(t) + t^2, \qquad y(0) = 0,$$

we can write the integral inequality

$$y(t) \leq \int_0^t \left(2s\, y(s) + s^2\right) ds.$$

This is of the general form

$$y(t) \leq \int_0^t a(s)\, y(s)\, ds + \int_0^t b(s)\, ds,$$

with $a(s) = 2s,\ b(s) = s^2$. The linear (differential) version of Grönwall gives that y is bounded above by the solution of the *equality*

$$z'(t) = 2t\, z(t) + t^2, \quad z(0) = 0,$$

which is exactly $z(t) = x(t)$ from part (a). Hence

$$y(t) \leq x(t) = e^{t^2} \int_0^t s^2 e^{-s^2}\, ds,$$

consistent with part (b). $\square$

Finite-Time Blowup

Problem

Consider the ODE

$$y' = y^3, \qquad y(0) = 1.$$

1. Solve explicitly and show that the solution blows up in finite time.
2. For the sequence $t_n = T - \frac{1}{n}$ where T is the blow-up time, compute $y(t_n)$.
3. Around each point $(t_n, y(t_n))$ apply the existence–uniqueness theorem in a rectangle and show that the guaranteed forward time of existence h_n satisfies $t_n + h_n < T$ for all n, with $h_n \to 0$ as $n \to \infty$.
4. Explain how this illustrates that the only obstruction to extending the solution beyond T is the unboundedness of $y(t)$.

Solution. We study

$$y' = y^3, \qquad y(0) = 1.$$

(a) Explicit solution and finite-time blow-up. Separate variables:

$$\frac{dy}{y^3} = dt.$$

Integrate:

$$\int y^{-3}\,dy = \int dt \quad \Longrightarrow \quad -\frac{1}{2y^2} = t + C.$$

Use $y(0) = 1$:

$$-\frac{1}{2 \cdot 1^2} = 0 + C \quad \Rightarrow \quad C = -\frac{1}{2}.$$

Thus

$$-\frac{1}{2y^2} = t - \frac{1}{2} \quad \Longrightarrow \quad \frac{1}{y^2} = 1 - 2t.$$

So

$$y(t) = \frac{1}{\sqrt{1-2t}}.$$

This solution exists only for $1 - 2t > 0$, i.e. for $t < \frac{1}{2}$. As $t \to T^- = \frac{1}{2}^-$,

$$y(t) = \frac{1}{\sqrt{1-2t}} \to +\infty.$$

Hence y blows up in finite time at $T = \frac{1}{2}$.

(b) Behavior of $y(t_n)$. Let $T = \frac{1}{2}$ and $t_n = T - \frac{1}{n} = \frac{1}{2} - \frac{1}{n}$. Then

$$1 - 2t_n = 1 - 2\Big(\frac{1}{2} - \frac{1}{n}\Big) = 1 - 1 + \frac{2}{n} = \frac{2}{n}.$$

Therefore

$$y(t_n) = \frac{1}{\sqrt{1-2t_n}} = \frac{1}{\sqrt{2/n}} = \sqrt{\frac{n}{2}},$$

so $y(t_n) \to \infty$ as $n \to \infty$.

(c) Local existence time around $(t_n, y(t_n))$. The right-hand side is $f(y) = y^3$, which is C^1 on $\mathbb{R}$ and locally Lipschitz.

The standard Picard–Lindelöf theorem says: for a given point (t_n, y_n), if we choose a rectangle

$$R_n = [\, t_n - a_n,\ t_n + a_n \,] \times [\, y_n - b_n,\ y_n + b_n \,],$$

then we have a solution guaranteed on at least a time interval of length

$$h_n = \min\left\{a_n,\ \frac{b_n}{M_n}\right\},$$

where

$$M_n := \sup_{(t,y)\in R_n} |f(y)| = \sup_{|y-y_n|\le b_n} |y^3|.$$

Since $y_n = y(t_n) = \sqrt{n/2} \to \infty$, if we fix some (say) $b_n = 1$ for simplicity, then

$$M_n \ge |y_n|^3 = \left(\sqrt{\tfrac{n}{2}}\right)^3 = \frac{n^{3/2}}{2\sqrt{2}} \to \infty.$$

Thus

$$\frac{b_n}{M_n} \le \frac{1}{|y_n|^3} \to 0,$$

so the guaranteed local existence time h_n satisfies $h_n \to 0$ as $n \to \infty$.

Also, because the maximal forward existence interval is $(0, T)$ with $T = \frac{1}{2}$, we must have

$$t_n + h_n < T = \frac{1}{2}$$

for each n: if $t_n + h_n \ge T$, the theorem would give an extension of the solution *past* T, contradicting blow-up there.

(d) Interpretation: obstruction is unboundedness. The extension theorem for ODEs says: if f is continuous and locally Lipschitz (as $f(y) = y^3$ is), then a solution can be extended beyond T as long as its graph stays in a compact subset of the domain (here, $(t, y) \in \mathbb{R}^2$). In particular, if $y(t)$ remained bounded on $[0, T)$, we could find a large bound R so that $|y(t)| \le R$ for all $t < T$, and apply the local existence theorem at $(T, y(T^-))$ to extend the solution.

But in this example, as $t \to T^- = \frac{1}{2}^-$ we have $y(t) \to \infty$, and the time-steps h_n from part (c) shrink to 0. This shows concretely that the *only* obstruction to extending the solution beyond T is that $y(t)$ becomes unbounded near T. $\square$

Problem

Determine whether solutions are global for any initial condition; justify your answer in each case.

1. $y' = \dfrac{y^2}{1+t^2}$.
2. $y' = \dfrac{y}{1+y^2}$.

Solution. We consider each equation separately and ask: is the solution global (defined for all $t \in \mathbb{R}$) for *every* initial condition?

(a) $y' = \dfrac{y^2}{1+t^2}$.

For simplicity take initial condition at $t = 0$: $y(0) = y_0$. The equation is separable:

$$\frac{dy}{y^2} = \frac{dt}{1+t^2}.$$

Integrate:

$$-\frac{1}{y} = \arctan t + C.$$

At $t = 0$,

$$-\frac{1}{y_0} = \arctan 0 + C = 0 + C \quad \Rightarrow \quad C = -\frac{1}{y_0}.$$

Thus

$$-\frac{1}{y} = \arctan t - \frac{1}{y_0} \quad \Longrightarrow \quad \frac{1}{y} = -\arctan t + \frac{1}{y_0}.$$

So

$$y(t) = \frac{1}{\frac{1}{y_0} - \arctan t}.$$

A blow-up occurs if the denominator becomes zero:

$$\frac{1}{y_0} - \arctan t = 0 \quad \Longleftrightarrow \quad \arctan t = \frac{1}{y_0}.$$

Since $\arctan t$ takes all values in $(-\frac{\pi}{2}, \frac{\pi}{2})$, there is a finite t solving this whenever

$$\frac{1}{y_0} \in \Big(-\frac{\pi}{2}, \frac{\pi}{2}\Big).$$

For example, if $y_0 = 1$, then $1/y_0 = 1 \in (-\pi/2, \pi/2)$ and blow-up occurs at $t = \tan(1)$.

Hence the solution is *not* global for all initial data: some initial conditions lead to finite-time blow-up (though others, e.g. $y_0 = 0$, do give global solutions).

(b) $y' = \dfrac{y}{1+y^2}$.

The right-hand side is $f(y) = \dfrac{y}{1+y^2}$. Note that

$$|f(y)| = \left|\frac{y}{1+y^2}\right| \le \max_{z\in\mathbb{R}} \left|\frac{z}{1+z^2}\right| = \frac{1}{2},$$

since the maximum occurs at $z = \pm 1$ with value $1/2$.

Also,

$$f'(y) = \frac{(1+y^2) - 2y^2}{(1+y^2)^2} = \frac{1-y^2}{(1+y^2)^2},$$

so $|f'(y)|$ is bounded for all y; thus f is globally Lipschitz.

Global Lipschitz continuity implies that for any initial condition $(t_0, y_0) \in \mathbb{R}^2$, the solution exists and is unique on all of $\mathbb{R}$, i.e. it is global. There is no finite-time blow-up because $|y'| \leq 1/2$ keeps the growth under control.

Summary: (a) not global for all initial data; (b) global for all initial data $\square$

Linear Stability

Problem

Consider

$$\begin{cases} x' = x(1-y), \\ y' = y(x-2). \end{cases}$$

1. Find all equilibria.
2. Compute the Jacobian at each equilibrium and find its eigenvalues.
3. Classify each equilibrium (saddle, node, spiral, center) for the linearized system.
4. Use the nonlinear linearization theorem (for hyperbolic equilibria) to state the nonlinear stability type of each equilibrium in the full system. (Where linearization is not conclusive, say so and explain why.)

Solution. We study

$$\begin{cases} x' = x(1-y), \\ y' = y(x-2). \end{cases}$$

(a) Equilibria. Equilibria satisfy

$$x(1-y) = 0, \qquad y(x-2) = 0.$$

From $x(1-y) = 0$ we get either $x = 0$ or $y = 1$. From $y(x-2) = 0$ we get either $y = 0$ or $x = 2$.

Combine:

- If $x = 0$, then $y(x-2) = 0$ becomes $y(-2) = 0 \Rightarrow y = 0$. So $(0,0)$.
- If $y = 1$, then $y(x-2) = 0$ becomes $1 \cdot (x-2) = 0 \Rightarrow x = 2$. So $(2,1)$.

Thus there are two equilibria:

$$(0,0), \quad (2,1).$$

(b) Jacobian and eigenvalues. Let $f(x,y) = x(1-y)$ and $g(x,y) = y(x-2)$. Then

$$Df(x,y) = \begin{pmatrix} \partial_x f & \partial_y f \\ \partial_x g & \partial_y g \end{pmatrix} = \begin{pmatrix} 1-y & -x \\ y & x-2 \end{pmatrix}.$$

At $(0,0)$:

$$Df(0,0) = \begin{pmatrix} 1 & 0 \\ 0 & -2 \end{pmatrix}.$$

Eigenvalues: $\lambda_1 = 1$, $\lambda_2 = -2$.

At $(2,1)$:

$$Df(2,1) = \begin{pmatrix} 0 & -2 \\ 1 & 0 \end{pmatrix}.$$

Characteristic polynomial:

$$\lambda^2 + 2 = 0 \quad \Longrightarrow \quad \lambda = \pm i\sqrt{2}.$$

(c) Linear classification.

- At $(0,0)$: eigenvalues 1 and -2 have opposite signs, so $(0,0)$ is a *saddle point* for the linearized system, hence linearly unstable.
- At $(2,1)$: eigenvalues are purely imaginary $\pm i\sqrt{2}$ (trace $= 0$, determinant > 0), so the linearization is a *center*: closed orbits in the linear approximation.

(d) Nonlinear stability via linearization theorem. The nonlinear system is C^1, so we can apply the linearization (Hartman–Grobman) theorem at *hyperbolic* equilibria, i.e. equilibria whose Jacobian has no eigenvalues with zero real part.

- $(0,0)$: eigenvalues are 1 and -2 (real parts > 0 and < 0), so $(0,0)$ is a hyperbolic saddle. The nonlinear system is topologically conjugate near $(0,0)$ to its linearization; hence $(0,0)$ is a *saddle* and *unstable* for the full nonlinear system.
- $(2,1)$: eigenvalues are purely imaginary $\pm i\sqrt{2}$, so this equilibrium is *not* hyperbolic. The linearization theorem does *not* apply. The linear system suggests a center, but the nonlinear terms could create a stable or unstable spiral or preserve a center. Using linearization alone, we cannot conclude the nonlinear stability type; we can only say that linearization is inconclusive at $(2,1)$.

$\square$

Lyapunov Functions

Problem

Consider
$$\begin{cases} x' = -x + 2y, \\ y' = -3x - y. \end{cases}$$

1. Show directly (by eigenvalues) that the origin is asymptotically stable.
2. Look for a Lyapunov function of the form
$$V(x, y) = ax^2 + bxy + cy^2$$
with $a > 0, c > 0$ and $4ac > b^2$.
   (i) Compute $\dot{V}$ along solutions and write it as a quadratic form in x, y.
   (ii) Choose a, b, c so that $\dot{V}$ is negative definite.
3. Conclude again that the origin is asymptotically stable by Lyapunov's direct method.

Solution. We consider
$$\begin{cases} x' = -x + 2y, \\ y' = -3x - y. \end{cases}$$
Write this as $X' = AX$ with
$$A = \begin{pmatrix} -1 & 2 \\ -3 & -1 \end{pmatrix}.$$

(a) Eigenvalues and asymptotic stability. The characteristic polynomial is
$$\det(A - \lambda I) = \det \begin{pmatrix} -1-\lambda & 2 \\ -3 & -1-\lambda \end{pmatrix} = (-1-\lambda)^2 - (-6) = (\lambda+1)^2 + 6.$$
So
$$(\lambda+1)^2 = -6 \quad \Longrightarrow \quad \lambda = -1 \pm i\sqrt{6}.$$
Both eigenvalues have real part $-1 < 0$. Thus for the linear system, the origin is an *asymptotically stable spiral*. In particular, it is asymptotically stable.

(b) Lyapunov function. We look at
$$V(x, y) = ax^2 + bxy + cy^2, \quad a > 0, \ c > 0, \ 4ac > b^2.$$
A convenient simple choice is
$$a = 1, \quad b = 0, \quad c = 1,$$
so
$$V(x, y) = x^2 + y^2.$$
This clearly satisfies $a > 0, c > 0$ and $4ac - b^2 = 4 > 0$, so V is positive definite.

Compute its derivative along solutions:

$$\dot{V} = \frac{d}{dt}(x^2 + y^2) = 2xx' + 2yy'.$$

Plug in the system:

$$\begin{aligned}\dot{V} &= 2x(-x + 2y) + 2y(-3x - y) \\ &= 2(-x^2 + 2xy - 3xy - y^2) \\ &= 2(-x^2 - xy - y^2) \\ &= -2(x^2 + xy + y^2).\end{aligned}$$

We check that $x^2 + xy + y^2$ is positive definite. Consider

$$x^2 + xy + y^2 = \frac{1}{2}\Big[(x + y)^2 + x^2 + y^2\Big].$$

Since $(x + y)^2 \geq 0$ and $x^2 + y^2 > 0$ for $(x, y) \neq (0, 0)$, we have

$$x^2 + xy + y^2 \geq \frac{1}{2}(x^2 + y^2) > 0 \quad \text{for } (x, y) \neq (0, 0).$$

Thus $x^2 + xy + y^2$ is positive definite, and hence

$$\dot{V} = -2(x^2 + xy + y^2)$$

is *negative definite*.
So with $V(x, y) = x^2 + y^2$, we have a positive definite Lyapunov function such that $\dot{V}$ is negative definite.

(c) Asymptotic stability by Lyapunov. Lyapunov's direct method says: if there is a neighborhood of the origin where

- V is positive definite, and
- $\dot{V}$ is negative definite,

then the equilibrium is *asymptotically stable*.

We have just shown both, and in fact these properties hold for all $(x, y) \in \mathbb{R}^2$. Therefore, the origin is globally asymptotically stable by Lyapunov's method, consistent with the eigenvalue analysis in part (a). □

Limit Sets

Problem

Consider the planar system in polar coordinates:

$$r' = r(1 - r^2), \qquad \theta' = 2.$$

1. Find all equilibria of the scalar ODE for $r(t)$ and classify their stability.
2. Describe the corresponding periodic orbits in the plane (circles of what radii?).
3. For initial data with $r_0 > 0$, describe the ω-limit set of the trajectory in $\mathbb{R}^2$.
4. For $0 < r_0 < 1$, describe the α-limit set; for $r_0 > 1$, describe the α-limit set.

Solution. We have the planar system in polar coordinates:

$$r' = r(1 - r^2), \qquad \theta' = 2.$$

(a) Equilibria and stability for r. Consider the scalar ODE

$$r' = r(1 - r^2).$$

Equilibria satisfy $r(1 - r^2) = 0$, so

$$r = 0, \quad r = \pm 1.$$

In planar polar coordinates we usually take $r \geq 0$, so the relevant equilibria in r are $r = 0$ and $r = 1$.

Compute $f(r) = r(1 - r^2)$, $f'(r) = 1 - 3r^2$.

- At $r = 0$: $f'(0) = 1 > 0 \Rightarrow r = 0$ is an *unstable* equilibrium for the scalar ODE.
- At $r = 1$: $f'(1) = 1 - 3 = -2 < 0 \Rightarrow r = 1$ is a *stable* equilibrium for the scalar ODE.

(b) Periodic orbits in the plane. In the full system, $\theta' = 2$ means $\theta(t) = 2t + \theta_0$ (uniform counterclockwise rotation).

- $r \equiv 0$ corresponds to the *equilibrium* at the origin $(0, 0)$.
- $r \equiv 1$ with $\theta(t) = 2t + \theta_0$ gives a solution that moves along the circle of radius 1 at constant angular speed. This is a periodic orbit:
$$\Gamma = \{(x, y) : x^2 + y^2 = 1\}.$$

Thus the nontrivial periodic orbit in the plane is the unit circle $r = 1$.

(c) ω-limit sets for $r_0 > 0$. For $r > 0$ the scalar equation $r' = r(1 - r^2)$ has the phase portrait:

$$0 \longrightarrow \cdots \longrightarrow 1,$$

i.e., for any $r_0 > 0$ with $r_0 \neq 1$ we have $r(t) \to 1$ as $t \to +\infty$ (solutions in $(0, \infty)$ are attracted to 1).

Thus, for any initial radius $r_0 > 0$ (and $r_0 \neq 0$), the radial coordinate tends to 1, while $\theta(t)$ grows linearly. In Cartesian coordinates, the trajectory spirals towards the unit circle.

The ω-limit set of such a trajectory is the entire unit circle:

$$\omega\big((r_0, \theta_0)\big) = \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 = 1\}.$$

(d) α-limit sets. We look at the behavior as $t \to -\infty$.

- If $0 < r_0 < 1$, then for increasing t, $r(t) \uparrow 1$, and for decreasing t (backwards in time), $r(t)$ decreases towards 0. Thus

  $$\lim_{t \to -\infty} r(t) = 0,$$

  and the trajectory approaches the origin in backward time. So the α-limit set is

  $$\alpha\big((r_0, \theta_0)\big) = \{(0, 0)\}, \qquad 0 < r_0 < 1.$$

- If $r_0 > 1$, then moving forward in time, $r(t) \downarrow 1$. Moving backward in time, the solution for r grows and tends to $+\infty$ in finite negative time. So the solution cannot be extended to arbitrarily large negative times; it "blows up" in backward time, leaving every compact subset of $\mathbb{R}^2$.

  Consequently, in the usual definition where $\alpha(x_0)$ is taken as the set of accumulation points as $t \to -\infty$, the α-limit set is empty:

  $$\alpha\big((r_0, \theta_0)\big) = \varnothing, \qquad r_0 > 1,$$

  because the trajectory is not defined for all sufficiently negative times and escapes to infinity instead of accumulating in the plane.

$\square$